# Namespace LlamaLogic.TuningDescriptions

## Classes

EnumTunableDescription

Represents the description of a tunable whose value is a member of an enumeration (a `TunableEnum` element) ( 🔒 )

FragmentReferenceTunableDescription

Represents a reference to a shared description (a `TdescFragTag` element), whose Class names a TuningDescriptionFragment ( 🔒 )

ListTunableDescription

Represents the description of a tunable that is a sequence of values described by Item (a `TunableList` element), including sets and mappings ( 🔒 )

ScalarTunableDescription

Represents the description of a single-valued tunable (a `Tunable` element) ( 🔒 )

SimDataGenerationException

Represents an exception thrown by SimDataGenerator when it cannot generate SimData for tuning

SimDataGenerator

Generates SimData resources from tuning the way Maxis' build does, using the tuning descriptions of the game version for which they are generated ( 🔒 )

TunableDescription

Represents the description of a tunable ( 🔒 )

TuningDescriptionAction

Represents a change Maxis recorded in a document's version history, identifying tunables by their Muid ( 🔒 )

TuningDescriptionClass

Represents a class described in a TuningDescriptionModule (a `Class` element): an enumeration, the holder of module tuning, or both ( 🔒 )

TuningDescriptionDocument

Represents a document of a TuningDescriptionSet: an instance, a module, or a fragment ( 🔒 )

TuningDescriptionEnumItem

Represents a member of an enumeration described by a TuningDescriptionClass (an `EnumItem` element) ( 🔒 )

TuningDescriptionFragment
: Represents a description shared by the FragmentReferenceTunableDescriptions that name it (a `TdescFrag` element), such as `TunableTestSet` ( 🔒 )

TuningDescriptionInstance
: Represents the description of a class of instance tuning (an `Instance` element), such as `Buff` ( 🔒 )

TuningDescriptionModule
: Represents the description of a Python module's tuning and enumerations (a `Module` element), such as `sims.sim_info_types` ( 🔒 )

TuningDescriptionSet
: Represents a set of tuning descriptions for a version of The Sims 4, as Maxis distributes it ( 🔒 )

TupleTunableDescription
: Represents the description of a tunable composed of named tunables (a `TunableTuple` element) ( 🔒 )

UnknownTunableDescription
: Represents a description in an element this library does not recognize, kept so that nothing in a document is lost ( 🔒 )

VariantTunableDescription
: Represents the description of a tunable whose value is one of several named options (a `TunableVariant` element) ( 🔒 )

## Enums

TuningDescriptionActionType
: The type of a TuningDescriptionAction

TuningDescriptionExportModes
: The destinations to which the game's build exports a tunable (its `export_modes` attribute)

# Class EnumTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable whose value is a member of an enumeration (a `TunableEnum` element) ( 🔒 )

```
public sealed class EnumTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← EnumTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### BinaryType

Gets the type the tunable's value takes when it is exported (its `binary_type` attribute)

```
public string? BinaryType { get; }
```

#### Property Value

string

### DynamicEntries

Gets the module tuning that adds members to the enumeration at run time (its `dynamic_entries` attribute), such as `tag.Tag._elements`

```
public string? DynamicEntries { get; }
```

### Property Value

string

### Remarks

Tuning descriptions do not contain these members; the game's module tuning does.

## InvalidEntries

Gets the members of the enumeration tuning may not use (its `invalid_entries` attribute), as written

```
public string? InvalidEntries { get; }
```

### Property Value

string

## StaticEntries

Gets the enumeration the tunable's values come from (its `static_entries` attribute), which Get Enum(string) finds

```
public string? StaticEntries { get; }
```

### Property Value

string

## Type

Gets the name of the enumeration (its `type` attribute)

```
public string? Type { get; }
```

## Property Value

string

# Class FragmentReferenceTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a reference to a shared description (a `TdescFragTag` element), whose Class names a TuningDescriptionFragment ( 🔒 )

```
public sealed class FragmentReferenceTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← FragmentReferenceTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Remarks

Resolve(TunableDescription) finds the description the reference stands for.

# Class ListTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable that is a sequence of values described by Item (a `TunableList` element), including sets and mappings ( 🔒 )

```
public sealed class ListTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← ListTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### AreEntriesUnique

Gets whether tuning may not repeat an entry (its `unique_entries` attribute)

```
public bool AreEntriesUnique { get; }
```

#### Property Value

bool

### IsMapping

Gets whether the list is a mapping, the items of which are pairs of the members named by MappingKey and MappingValue

```
public bool IsMapping { get; }
```

### Property Value

bool

## Item

Gets the description of the list's items, or null if the description has none

```
public TunableDescription? Item { get; }
```

### Property Value

TunableDescription

## MappingClass

Gets the name of the class of the mapping's pairs (its `mapping_class` attribute)

```
public string? MappingClass { get; }
```

### Property Value

string

## MappingKey

Gets the name of the member of Item that is the key of each pair (its `mapping_key` attribute)

```
public string? MappingKey { get; }
```

### Property Value

string

# MappingValue

Gets the name of the member of Item that is the value of each pair (its `mapping_value` attribute)

```
public string? MappingValue { get; }
```

## Property Value

string

# Class ScalarTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a single-valued tunable (a `Tunable` element) ( 🔒 )

```
public sealed class ScalarTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← ScalarTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### AllowNone

Gets whether the tunable may be `None` (its `allow_none` attribute), or null if the description does not say

```
public bool? AllowNone { get; }
```

#### Property Value

bool?

### BinaryType

Gets the type the tunable's value takes when it is exported (its `binary_type` attribute)

```
public string? BinaryType { get; }
```

### Property Value

string

## Type

Gets the type of the tunable's value (its `type` attribute): a Python type such as `int`, `ResourceKey`, or the instance type of the tuning it references

```
public string? Type { get; }
```

### Property Value

string

# Class SimDataGenerationException

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents an exception thrown by SimDataGenerator when it cannot generate SimData for tuning

```
public sealed class SimDataGenerationException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← SimDataGenerationException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.ReferenceEquals(object, object)

## Constructors

### SimDataGenerationException()

Initializes a new SimDataGenerationException

```
public SimDataGenerationException()
```

### SimDataGenerationException(Exception?)

Initializes a new SimDataGenerationException with the specified `innerException`

```
public SimDataGenerationException(Exception? innerException)
```

### Parameters

`innerException` Exception

## SimDataGenerationException(string)

Initializes a new SimDataGenerationException with the specified `message`

```
public SimDataGenerationException(string message)
```

### Parameters

`message` string

## SimDataGenerationException(string, Exception?)

Initializes a new SimDataGenerationException with the specified `message` and `innerException`

```
public SimDataGenerationException(string message, Exception? innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class SimDataGenerator

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Generates SimData resources from tuning the way Maxis' build does, using the tuning descriptions of the game version for which they are generated ( 🔒 )

```
public sealed class SimDataGenerator
```

**Inheritance**

object ← SimDataGenerator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Which tunables become columns, the columns' types, and the values written for tuning that omits a tunable all come from Descriptions.

Some of what Maxis' build knows is not in tuning descriptions, so the generator learns it:

- AddTuning(XDocument) and AddTuning(CombinedTuningModel) learn the entries of dynamic enumerations (such as tags defined by module tuning) and the instances of tuning that is referenced by name
- AddLayouts(SimDataModel) learns the column order of schemas the game reads with a layout of its own (such as `Client_*` and `Renderer.*` schemas), whose columns are otherwise ordered by name
- AddResourceKeyConversion(ResourceType, uint, ResourceType, uint) learns resource key conversions (Maxis' build converts the keys of resources it also converts, such as PNG images, which tuning written for mods should not have converted)

For the game's own SimData, the game's combined tuning and SimData are the sources of what the generator needs to learn.

# Constructors

## SimDataGenerator(TuningDescriptionSet)

Initializes a new SimDataGenerator

```
public SimDataGenerator(TuningDescriptionSet descriptions)
```

### Parameters

`descriptions` TuningDescriptionSet

The tuning descriptions of the game version for which SimData will be generated

## Properties

### Descriptions

Gets the tuning descriptions of the game version for which SimData is generated

```
public TuningDescriptionSet Descriptions { get; }
```

### Property Value

TuningDescriptionSet

## Methods

### AddLayouts(SimDataModel)

Learns the column order of each schema in the specified `simData` whose columns are not ordered by name

```
public void AddLayouts(SimDataModel simData)
```

### Parameters

`simData` SimDataModel

### Remarks

When generating an object of a schema with the same name and the same columns, the generator orders the columns as they are ordered in `simData`.

## AddResourceKeyConversion(ResourceType, uint, ResourceType, uint)

Learns that resource keys of the specified `fromType` and `fromGroup` in tuning are written to SimData with the specified `toType` and `toGroup`

```
public void AddResourceKeyConversion(ResourceType fromType, uint fromGroup, ResourceType toType, uint toGroup)
```

### Parameters

`fromType` ResourceType

`fromGroup` uint

`toType` ResourceType

`toGroup` uint

## AddTuning(CombinedTuningModel)

Learns the entries of dynamic enumerations and the names of instances from each tuning in the specified `combinedTuning`

```
public void AddTuning(CombinedTuningModel combinedTuning)
```

### Parameters

`combinedTuning` CombinedTuningModel

## AddTuning(XDocument)

Learns the entries of dynamic enumerations the specified `tuning` defines (if it is module tuning) and the name of its instance

```
public void AddTuning(XDocument tuning)
```

### Parameters

`tuning` XDocument

### Exceptions

FormatException

The instance of `tuning` is not a number

## Generate(XDocument)

Generates the SimData for the specified `tuning`

```
public SimDataModel Generate(XDocument tuning)
```

### Parameters

`tuning` XDocument

Instance tuning (an `I` element) or module tuning (an `M` element)

### Returns

SimDataModel

The SimData, with one instance for instance tuning or one instance for each class of module tuning that has exported tunables

### Remarks

The Pack of the SimData is `0`, the base game's; set it for SimData that belongs to a pack.

### Exceptions

SimDataGenerationException

`tuning` cannot be generated with Descriptions and what the generator has learned

# Class TunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable ( 🔒 )

```
public abstract class TunableDescription
```

**Inheritance**

object ← TunableDescription

**Derived**

EnumTunableDescription, FragmentReferenceTunableDescription, ListTunableDescription, ScalarTunableDescription, TupleTunableDescription, UnknownTunableDescription, VariantTunableDescription

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Every attribute of the description's XML element is available in Attributes, including those this library does not give properties of their own.

# Properties

## Attributes

Gets all of the description's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

### Property Value

IReadOnlyDictionary<string, string>

## Class

Gets the Python class of the tunable (its `class` attribute)

```
public string? Class { get; }
```

### Property Value

string

## Default

Gets the tunable's default value, as written (its `default` attribute)

```
public string? Default { get; }
```

### Property Value

string

## Description

Gets the description of the tunable for tuners (its `description` attribute)

```
public string? Description { get; }
```

### Property Value

string

## Display

Gets the display name of the tunable (its `display` attribute)

```
public string? Display { get; }
```

### Property Value

string

## ExportModes

Gets the destinations to which the tunable is exported (its `export_modes` attribute), or null if the description does not specify them

```
public TuningDescriptionExportModes? ExportModes { get; }
```

### Property Value

TuningDescriptionExportModes?

## Group

Gets the group of the tunable in tuning editors (its `group` attribute)

```
public string? Group { get; }
```

### Property Value

string

## IsDeprecated

Gets whether the tunable is deprecated (its `Deprecated` attribute)

```
public bool IsDeprecated { get; }
```

### Property Value

bool

## Members

Gets the descriptions nested in this one, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

### Property Value

IReadOnlyList<TunableDescription>

## Muid

Gets the unique identifier Maxis assigns the tunable (its `muid` attribute)

```
public ulong? Muid { get; }
```

### Property Value

ulong?

## Name

Gets the name of the tunable (its `name` attribute), or null for the item of a list

```
public string? Name { get; }
```

### Property Value

string

# Methods

## GetMember(string)

Gets the nested description with the specified `name`, or null if there is none

```
public TunableDescription? GetMember(string name)
```

## Parameters

`name` string

## Returns

TunableDescription

# Class TuningDescriptionAction

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a change Maxis recorded in a document's version history, identifying tunables by their Muid ( 🔒 )

```
public sealed class TuningDescriptionAction
```

**Inheritance**

object ← TuningDescriptionAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Attributes

Gets all of the action's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

### Property Value

IReadOnlyDictionary <string, string>

## NewMuid

Gets the MUID that replaces OldMuid (its `new_muid` attribute)

```
public ulong? NewMuid { get; }
```

### Property Value

ulong?

## OldMuid

Gets the MUID the action removes or renames (its `old_muid` attribute)

```
public ulong? OldMuid { get; }
```

### Property Value

ulong?

## SourceTypeValue

Gets the option value the action changes from (its `source_type_value` attribute)

```
public string? SourceTypeValue { get; }
```

### Property Value

string

## TargetMuid

Gets the MUID of the tunable the action changes (its `target_muid` attribute)

```
public ulong? TargetMuid { get; }
```

### Property Value

ulong?

## TargetTypeValue

Gets the option value the action changes to (its `target_type_value` attribute)

```
public string? TargetTypeValue { get; }
```

### Property Value

string

## Type

Gets the type of the action

```
public TuningDescriptionActionType Type { get; }
```

### Property Value

TuningDescriptionActionType

## TypeName

Gets the type of the action as written (its `type` attribute)

```
public string TypeName { get; }
```

### Property Value

string

# Enum TuningDescriptionActionType

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

The type of a TuningDescriptionAction

```
public enum TuningDescriptionActionType
```

## Fields

`ChangeInheritance = 4`

A class's inheritance changed (`kChangeInheritance`)

`Remove = 1`

A tunable was removed (`kRemove`)

`Rename = 2`

A tunable was renamed (`kRename`)

`Unknown = 0`

A type this library does not recognize (see TypeName)

`UpdateVariantTypeField = 3`

The value naming a variant's option changed (`kUpdateVariantTypeField`)

# Class TuningDescriptionClass

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a class described in a TuningDescriptionModule (a `Class` element): an enumeration, the holder of module tuning, or both ( 🔒 )

```
public sealed class TuningDescriptionClass
```

**Inheritance**

object ← TuningDescriptionClass

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Attributes

Gets all of the class's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

### Property Value

IReadOnlyDictionary<string, string>

## Classes

Gets the classes nested in this one, in document order

```
public IReadOnlyList<TuningDescriptionClass> Classes { get; }
```

### Property Value

IReadOnlyList<TuningDescriptionClass>

## EnumItems

Gets the members of the enumeration the class is, in document order

```
public IReadOnlyList<TuningDescriptionEnumItem> EnumItems { get; }
```

### Property Value

IReadOnlyList<TuningDescriptionEnumItem>

## IsBitFlagEnum

Gets whether the class is an enumeration of bit flags (its `enum_bit_flag` attribute)

```
public bool IsBitFlagEnum { get; }
```

### Property Value

bool

## IsEnum

Gets whether the class is an enumeration

```
public bool IsEnum { get; }
```

### Property Value

bool

## IsLockedEnum

Gets whether the enumeration's members are fixed (its `enum_locked` attribute)

```
public bool IsLockedEnum { get; }
```

### Property Value

bool

## Members

Gets the module tuning the class declares, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

### Property Value

IReadOnlyList<TunableDescription>

## Module

Gets the module in which the class is described

```
public TuningDescriptionModule Module { get; }
```

### Property Value

TuningDescriptionModule

## Muid

Gets the unique identifier Maxis assigns the class (its `muid` attribute)

```
public ulong? Muid { get; }
```

### Property Value

ulong?

## Name

Gets the name of the class (its `name` attribute)

```
public string Name { get; }
```

### Property Value

string

## Parent

Gets the class in which this one is nested, or null if it is not nested

```
public TuningDescriptionClass? Parent { get; }
```

### Property Value

TuningDescriptionClass

## StaticEntriesName

Gets the name by which a StaticEntries attribute refers to the class, such as `sims-sim_info_types.Age`

```
public string StaticEntriesName { get; }
```

### Property Value

string

# Methods

## GetClass(string)

Gets the nested class with the specified `name`, or null if there is none

```
public TuningDescriptionClass? GetClass(string name)
```

### Parameters

`name` string

### Returns

TuningDescriptionClass

## GetEnumValue(string)

Gets the value of the enumeration member with the specified `name`, or null if there is none

```
public long? GetEnumValue(string name)
```

### Parameters

`name` string

### Returns

long?

# Class TuningDescriptionDocument

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a document of a TuningDescriptionSet: an instance, a module, or a fragment ( 🔒 )

```
public abstract class TuningDescriptionDocument
```

**Inheritance**

object ← TuningDescriptionDocument

**Derived**

TuningDescriptionFragment, TuningDescriptionInstance, TuningDescriptionModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Attributes

Gets all of the attributes of the document's root element, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

#### Property Value

IReadOnlyDictionary<string, string>

### Muid

Gets the unique identifier Maxis assigns the document (its `muid` attribute)

```
public ulong? Muid { get; }
```

### Property Value

ulong?

## Path

Gets the path of the document in its set, with `/` separating directories

```
public string Path { get; }
```

### Property Value

string

## Set

Gets the set to which the document belongs

```
public TuningDescriptionSet Set { get; }
```

### Property Value

TuningDescriptionSet

## VersionHistory

Gets the changes Maxis recorded for the document, one list per `ActionList` element, in document order

```
public IReadOnlyList<IReadOnlyList<TuningDescriptionAction>> VersionHistory { get; }
```

### Property Value

IReadOnlyList<IReadOnlyList<TuningDescriptionAction>>

# Class TuningDescriptionEnumItem

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a member of an enumeration described by a TuningDescriptionClass (an `EnumItem` element) ( 🔒 )

```
public sealed class TuningDescriptionEnumItem
```

**Inheritance**

object ← TuningDescriptionEnumItem

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Attributes

Gets all of the member's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

### Property Value

IReadOnlyDictionary<string, string>

## Muid

Gets the unique identifier Maxis assigns the member (its `muid` attribute)

```
public ulong? Muid { get; }
```

### Property Value

ulong?

## Name

Gets the name of the member (its `name` attribute)

```
public string Name { get; }
```

### Property Value

string

## Value

Gets the value of the member (its `enum_value` attribute)

```
public long Value { get; }
```

### Property Value

long

# Enum TuningDescriptionExportModes

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

The destinations to which the game's build exports a tunable (its `export_modes` attribute)

```
[Flags]
public enum TuningDescriptionExportModes
```

## Fields

`ClientBinary = 1`

The tunable is exported to `SimData` resources (`client_binary`)

`None = 0`

No destinations

`ServerBinary = 2`

The tunable is exported to the binary combined tuning (`server_binary`)

`ServerXml = 4`

The tunable is exported to tuning XML (`server_xml`)

# Class TuningDescriptionFragment

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a description shared by the FragmentReferenceTunableDescriptions that name it (a `TdescFrag` element), such as `TunableTestSet` ( 🔒 )

```
public sealed class TuningDescriptionFragment : TuningDescriptionDocument
```

**Inheritance**

object ← TuningDescriptionDocument ← TuningDescriptionFragment

**Inherited Members**

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Class

Gets the name by which references refer to the fragment (its description's `class` attribute)

```
public string Class { get; }
```

#### Property Value

string

### Description

Gets the description the fragment shares

```
public TunableDescription Description { get; }
```

## Property Value

TunableDescription

# Class TuningDescriptionInstance

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a class of instance tuning (an `Instance` element), such as `Buff` ( 🔒 )

```
public sealed class TuningDescriptionInstance : TuningDescriptionDocument
```

**Inheritance**

object ← TuningDescriptionDocument ← TuningDescriptionInstance

**Inherited Members**

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Members holds only the tunables the class declares; GetEffectiveMembers() adds those it inherits.

# Properties

## Class

Gets the Python class of the instance tuning (its `class` attribute)

```
public string Class { get; }
```

### Property Value

string

## DeletedMembers

Gets the names of the inherited tunables the class removes (its `Deleted` elements)

```
public IReadOnlyList<string> DeletedMembers { get; }
```

### Property Value

IReadOnlyList <string>

## Description

Gets the description of the class for tuners (its `description` attribute)

```
public string? Description { get; }
```

### Property Value

string

## ExportModes

Gets the destinations to which the whole class is exported (its `export_modes` attribute), or null if the description does not specify them

```
public TuningDescriptionExportModes? ExportModes { get; }
```

### Property Value

TuningDescriptionExportModes?

## InstanceType

Gets the type of instance tuning the class is (its `instance_type` attribute), such as `buff`

```
public string? InstanceType { get; }
```

### Property Value

string

## Members

Gets the tunables the class declares, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

### Property Value

IReadOnlyList <TunableDescription>

## Module

Gets the Python module that defines the class (its `module` attribute)

```
public string? Module { get; }
```

### Property Value

string

## Parents

Gets the classes from which the class inherits (its `parents` attribute), in declaration order

```
public IReadOnlyList<string> Parents { get; }
```

### Property Value

IReadOnlyList <string >

# Methods

## GetEffectiveMembers()

Gets the tunables of the class, including those it inherits and excluding those it deletes

```
public IReadOnlyList<TunableDescription> GetEffectiveMembers()
```

### Returns

IReadOnlyList<TunableDescription>

### Remarks

Each tunable is the description from the first class in GetMethodResolutionOrder() that declares or deletes it, as in Python. The tunables are in the order the game collects them: walking from the most basic class to this one, each class first removes the tunables it deletes and then adds or replaces those it declares, a newly added tunable going to the end.

### Exceptions

InvalidDataException

The class's inheritance cannot be resolved (see GetMethodResolutionOrder())

## GetMember(string)

Gets the tunable the class declares with the specified `name`, or null if it declares none

```
public TunableDescription? GetMember(string name)
```

### Parameters

`name` string

### Returns

TunableDescription

## GetMethodResolutionOrder()

Gets this class followed by the classes from which it inherits, in the order Python searches them (the C3 linearization of Parents)

```
public IReadOnlyList<TuningDescriptionInstance> GetMethodResolutionOrder()
```

## Returns

IReadOnlyList <TuningDescriptionInstance>

## Exceptions

InvalidDataException

A parent is not in the set, the inheritance is circular, or the parents cannot be linearized

# Class TuningDescriptionModule

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a Python module's tuning and enumerations (a `Module` element), such as `sims.sim_info_types` ( 🔒 )

```
public sealed class TuningDescriptionModule : TuningDescriptionDocument
```

**Inheritance**

object ← TuningDescriptionDocument ← TuningDescriptionModule

**Inherited Members**

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Classes

Gets the module's classes, in document order

```
public IReadOnlyList<TuningDescriptionClass> Classes { get; }
```

### Property Value

IReadOnlyList<TuningDescriptionClass>

## Name

Gets the name of the module (its `name` attribute)

```
public string Name { get; }
```

### Property Value

string

## Methods

### GetClass(string)

Gets the class of the module with the specified `name`, or null if there is none

```
public TuningDescriptionClass? GetClass(string name)
```

### Parameters

`name` string

### Returns

TuningDescriptionClass

# Class TuningDescriptionSet

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a set of tuning descriptions for a version of The Sims 4, as Maxis distributes it ( 🔒 )

```
public sealed class TuningDescriptionSet : IDisposable
```

**Inheritance**

object ← TuningDescriptionSet

**Implements**

IDisposable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

A set is indexed by the paths of its documents when it is opened: instances are `*/Descriptions/{class}.tdesc`, modules are `Tuning/{module with . replaced by -}.tdesc`, and fragments are `TdescFrag/{class}.tdescfrag`. A document is read the first time it is requested, and it may be read again once nothing refers to it any longer, so requesting a document twice does not necessarily return the same object.

A set's members are safe to use from multiple threads.

# Properties

## FragmentClasses

Gets the names of the classes of the set's fragments

```
public IReadOnlyCollection<string> FragmentClasses { get; }
```

### Property Value

IReadOnlyCollection<string>

## GameVersion

Gets the version of the game the set describes, if it was opened from a file named as Maxis names them

```
public Version? GameVersion { get; }
```

### Property Value

Version

## InstanceClasses

Gets the names of the classes of the set's instances

```
public IReadOnlyCollection<string> InstanceClasses { get; }
```

### Property Value

IReadOnlyCollection<string>

## InstanceTypes

Gets the types of instance tuning the set's instances are (their `instance_type` attributes), such as `buff`

```
public IReadOnlyCollection<string> InstanceTypes { get; }
```

### Property Value

IReadOnlyCollection<string>

### Remarks

The types are read from the beginning of each instance's document, which is much faster than reading every instance.

### Exceptions

InvalidDataException

An instance's document is malformed

## ModuleNames

Gets the names of the set's modules

```
public IReadOnlyCollection<string> ModuleNames { get; }
```

### Property Value

IReadOnlyCollection<string>

# Methods

## Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

## FromDirectory(string)

Opens the set of tuning descriptions extracted to the directory at the specified `path` ( )

```
public static TuningDescriptionSet FromDirectory(string path)
```

### Parameters

`path` string

### Returns

TuningDescriptionSet

### Exceptions

DirectoryNotFoundException

There is no directory at `path`

## FromZip(Stream)

Opens the ZIP archive of tuning descriptions in the specified `stream`, which must remain open and seekable until the set is disposed ( 🗑 )

```
public static TuningDescriptionSet FromZip(Stream stream)
```

### Parameters

`stream` Stream

### Returns

TuningDescriptionSet

## FromZip(string)

Opens the ZIP archive of tuning descriptions at the specified `path` ( 🗑 )

```
public static TuningDescriptionSet FromZip(string path)
```

### Parameters

`path` string

### Returns

TuningDescriptionSet

### Remarks

When the name of the file is that of a Maxis distribution (such as `1.128.90.1030-TDESC.zip`), Game Version is the version it names.

## GetEnum(string)

Gets the enumeration to which an StaticEntries attribute refers, such as `sims-sim_info_types.Age`, or null if the set does not describe it

```
public TuningDescriptionClass? GetEnum(string staticEntries)
```

### Parameters

`staticEntries` string

### Returns

TuningDescriptionClass

### Exceptions

InvalidDataException

The module that would describe the enumeration is malformed

## GetFragment(string)

Gets the fragment with the specified `className`, or null if the set does not have it

```
public TuningDescriptionFragment? GetFragment(string className)
```

### Parameters

`className` string

### Returns

TuningDescriptionFragment

### Exceptions

InvalidDataException

The fragment's document is malformed or describes a different fragment

## GetInstance(string)

Gets the instance with the specified `className`, or null if the set does not have it

```
public TuningDescriptionInstance? GetInstance(string className)
```

### Parameters

`className` string

### Returns

TuningDescriptionInstance

### Exceptions

InvalidDataException

The instance's document is malformed or describes a different class

## GetModule(string)

Gets the module with the specified `name`, such as `sims.sim_info_types`, or null if the set does not have it

```
public TuningDescriptionModule? GetModule(string name)
```

### Parameters

`name` string

### Returns

TuningDescriptionModule

### Exceptions

InvalidDataException

The module's document is malformed or describes a different module

# Resolve(TunableDescription)

Gets the description the specified `description` stands for: the description of the fragment it refers to if it is a FragmentReferenceTunableDescription, and otherwise itself

```
public TunableDescription Resolve(TunableDescription description)
```

## Parameters

`description` TunableDescription

## Returns

TunableDescription

## Exceptions

InvalidDataException

The set does not have the fragment to which `description` refers

# Class TupleTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable composed of named tunables (a `TunableTuple` element) ( 🔒 )

```
public sealed class TupleTunableDescription : TunableDescription
```

### Inheritance

object ← TunableDescription ← TupleTunableDescription

### Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Class UnknownTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a description in an element this library does not recognize, kept so that nothing in a document is lost ( 🔒 )

```
public sealed class UnknownTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← UnknownTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ElementName

Gets the name of the description's XML element

```
public string ElementName { get; }
```

### Property Value

string

# Class VariantTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable whose value is one of several named options (a `TunableVariant` element) ( 🔒 )

```
public sealed class VariantTunableDescription : TunableDescription
```

**Inheritance**

object ← TunableDescription ← VariantTunableDescription

**Inherited Members**

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Default is the name of the option tuning gets when it does not choose one.

# Properties

## Options

Gets the options of the variant, in document order

```
public IReadOnlyList<TunableDescription> Options { get; }
```

### Property Value

IReadOnlyList<TunableDescription>

# Methods

## GetOption(string)

Gets the option with the specified `name`, or null if there is none

```
public TunableDescription? GetOption(string name)
```

### Parameters

`name` string

### Returns

TunableDescription